

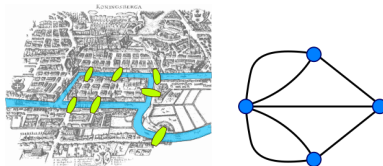
# CS 105: Department Introductory Course on Discrete Structures

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## Lecture 19 – An Introduction to Graph Theory

## Defining graphs



### Definition

A simple graph  $G$  is a pair  $(V, E)$  of a set of vertices/nodes  $V$  and edges  $E$  between unordered pairs of vertices called end-points:  $e = vu$  means  $e$  is an edge between  $v$  and  $u$  ( $u \neq v$ ).

### General Definition

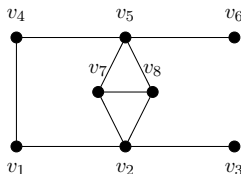
A graph  $G$  is a triple  $V, E, R$  where  $V$  is a set of vertices,  $E$  is a set of edges and  $R \subseteq E \times V \times V$  is a relation.

We will consider **only** finite graphs (i.e.,  $|V|, |E|$  are finite) and **often** simple graphs. **Also, we assume**  $|V| > 0$ .

## Basic terminology

- ▶ The **degree**  $d(v)$  of a vertex  $v$  (in an undirected loopless graph) is the number of edges incident to it, i.e.,  $|\{vw \in E \mid w \in V\}|$ . A vertex of degree 0 is called an **isolated vertex**.
- ▶ A **walk** is a sequence of vertices  $v_1, \dots, v_k$  such that  $\forall i \in \{1, \dots, k-1\}$ ,  $(v_i, v_{i+1}) \in E$ . **Vertices**  $v_1$  **and**  $v_k$  **are called** **end-points** and others are called **internal vertices**.
- ▶ A walk is called **closed** if it starts and ends with the same vertex, i.e., its endpoints are the same.
- ▶ A graph is **connected** if there is a walk between every two vertices.
- ▶ The **length** of a walk is the number of edges in it.

## Basic terminology: Post Lunch Quiz Part 1

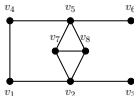


1. Give examples for each of the following in the above graph:
  - 1.1 All vertices of degree 3
  - 1.2 A walk of length 5
  - 1.3 A closed walk of length 6
2. True or False: (a) If a graph is not connected, it must have an isolated vertex.  
(b) A graph is connected iff some vertex has an edge to every other vertex.

## Post Lunch Quiz: Part 2

### More definitions

- ▶ A **path** is a walk in which no vertex is repeated.
- ▶ A **cycle** is a (non-empty) closed walk whose internal vertices are all distinct from each other and from the end-point **and edges are also distinct**.
- ▶ A path  $P$  is said to be **maximal** in  $G$  if it is not contained in any longer path.



- Give examples for each of the following in the above graph:
  - A path of length 5 and a cycle of length 6.
  - A maximal path that is not maximum.
- \*Every closed walk whose length is odd contains an cycle whose length is odd. (contains means vertices and edges of path occur as a sublist of walk, in cyclic order but not necessarily consecutive.)

## Some simple types of Graphs

- ▶ We have already seen some: connected graphs, complete graphs.
- ▶ paths, cycles.
- ▶ Are there other interesting classes of graphs?

## Bipartite graphs

### Definition

A graph is called **bipartite**, if the vertices of the graph can be partitioned into  $V = X \cup Y$ ,  $X \cap Y = \emptyset$  s.t.,  $\forall e = (u, v) \in E$ ,

- ▶ either  $u \in X$  and  $v \in Y$
- ▶ or  $v \in X$  and  $u \in Y$

Example:  $m$  jobs and  $n$  people,  $k$  courses and  $\ell$  students.

- ▶ How can we check if a graph is bipartite?
- ▶ Can we characterize bipartite graphs?

### Exercises

1. What is largest  $n$  s.t  $n$ -vertex complete graph is bipartite?
2. Can a  $n$ -vertex bipartite graph have  $n - 1$  degree vertex?
3. A graph is bipartite iff we can color vertices with 2 colors s.t no 2 adjacent vertices have same color.

## Characterizing bipartite graphs using cycles.

- ▶ Recall: A path or a cycle has length  $n$  if the number of edges in it is  $n$ .
- ▶ A path (or cycle) is call odd (or even) if its length is odd (or even, respectively).

Exercise: Prove or Disprove:

Every closed odd walk contains an odd cycle.



## Characterizing bipartite graphs using cycles.

**Exercise:** Prove or Disprove:

Every closed odd walk contains an odd cycle.

Proof: By induction on the length of the given closed odd walk.

## Characterizing bipartite graphs using cycles.

### Lemma

Every closed odd walk contains an odd cycle.

### Theorem, Konig, 1936

A graph is bipartite iff it has no odd cycle.

Proof:

- ▶ (  $\implies$  ) direction is easy.
  - ▶ Let  $G$  be bipartite with  $(V = X \cup Y)$ . Then, every walk in  $G$  alternates between  $X, Y$ .
- $\implies$  if we start from  $X$ , each return to  $X$  can only happen after an even number of steps.
- $\implies$   $G$  has no odd cycles.

## Characterizing bipartite graphs using cycles.

### Lemma

Every closed odd walk contains an odd cycle.

### Theorem, König, 1936

A graph is bipartite iff it has no odd cycle.

Proof:

- ▶ ( $\Leftarrow$ ) Suppose  $G$  has no odd cycle, then let us construct the bipartition. Wlog assume  $G$  is connected.
- ▶ Let  $u \in V$ . Break  $V$  into
$$X = \{v \in V \mid \text{length of shortest path } P_{uv} \text{ from } u \text{ to } v \text{ is even}\},$$
$$Y = \{v \in V \mid \text{length of shortest path } P_{uv} \text{ from } u \text{ to } v \text{ is odd}\},$$
- ▶ If there is an edge  $vv'$  between two vertices of  $X$  or two vertices of  $Y$ , this creates a closed odd walk:  $uP_{uv}vv'P_{v'u}u$ .
- ▶ By Lemma, it must contain an odd cycle: contradiction.
- ▶ This along with  $X \cap Y = \emptyset$  and  $X \cup Y = V$ , implies  $X, Y$  is a bipartition.  $\square$